

Deep Architecture Design Review

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Scope: Architecture Self-Critique, Modular Boundaries, WhatsApp UX/System, Mobile Strategy & Scaling Roadmap

Phase: Deep Architecture Analysis & System Blueprint (Zero Code Modification)

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EXECUTIVE DESIGN REVIEW SUMMARY:

This document presents the Phase 2 Deep Architecture Review for SmartLegal AI. It rigorously critiques initial architectural assumptions, simplifies components for maximum team efficiency, defines strict Domain & Module boundaries, designs complete product & technical specs for the WhatsApp Bot and Mobile App (Capacitor), and establishes an explicit zero-cost initial launch model scaling to enterprise capacity.

5. Mobile Architecture (Capacitor Strategy)

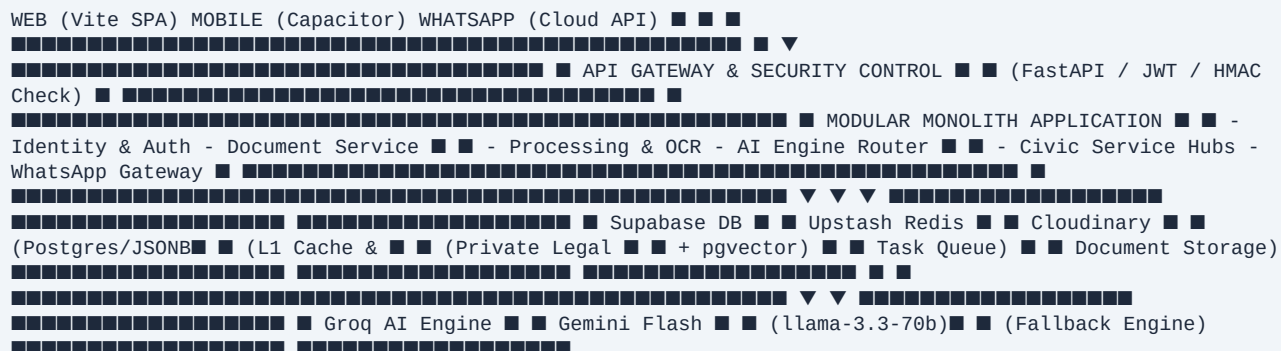
The mobile platform utilizes a hybrid Capacitor wrapper around the single React 18 codebase, sharing 95% of frontend code.

Feature Area	Shared Web/Mobile Implementation	Platform Native Plugin
Authentication	Redux `authSlice` + JWT API Client	`@capacitor-community/secure-storage` (Keychain/Keystore)
Document Picker	Dropzone UI & File validation	`@capacitor/file-picker` for native PDF selection
Notifications	In-app notification banner	`@capacitor/push-notifications` (FCM / APNs)
Deep Linking	React Router splat routes	`@capacitor/app` deep link listener (`smartlegal://`)

6. Deployment Stages & Cost Scaling Roadmap

Scaling Stage	Target Traffic	Infrastructure Stack	Est. Monthly Cost
Stage 0 (Free)	0 - 1,000 MAU	Vercel + Render Free + Supabase Free + Groq Free	■0 / \$0
Stage 1 (Small)	1,000 - 10,000 MAU	Render Starter (\$7) + Supabase Pro (\$25) + Upstash (\$3)	~\$35 / mo
Stage 2 (Growing)	10,000 - 50,000 MAU	Railway/ECS (\$50) + Supabase Pro Compute (\$60) + Redis (\$30) + Groq Commercial (\$50)	~\$220 / mo
Stage 3 (Scale)	50,000+ MAU	AWS Multi-AZ ECS + Aurora Postgres + Redis Cluster + Pinecone Vector + Dedicated OCR Cluster	~\$1,200+ / mo

7. Final Target System Architecture Blueprint



Report End — SmartLegal AI Deep Design Strategy Document